Two-Loop in $\frac{\lambda}{4!}\phi^4$ Theory

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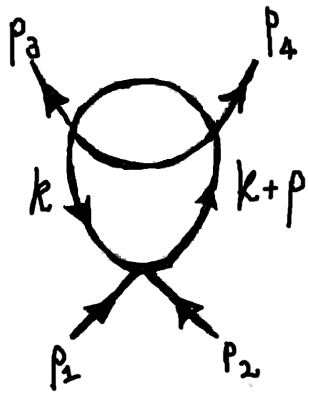
Correction to yanxi@tcd.ie or DM me via messenger.

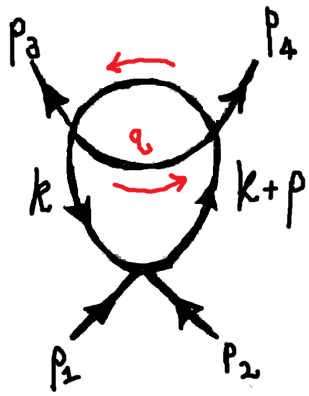
(a) Write down the expression for the following 2-loop Feynman diagram $V_{[\text{2-loop}]}$ in $\frac{1}{4}\lambda\phi^4$ theory



in terms of the integral over undetermined loop momentum k and 1-loop 4-point sub-diagram $V_{[\text{1-loop}]}$ (with outgoing momenta $[p_3, p_4]$ and incoming momenta $[-k, k+p]$).

Answer:

Once again, let's label the diagram, we simply add label q to complete the second loop:



Via conservation of momentum, the unlabeled red arrow has momentum

$$q + p_3 + k$$

Then use Feynman rule for ϕ^4 theory, we can write down the integral

$$i\mathcal{M} = (-i\lambda)^3 \int \frac{d^dk}{(2\pi)^d} \int \frac{d^dq}{(2\pi)^d} \frac{i}{k^2-m^2} \frac{i}{(k+p)^2-m^2} \frac{i}{q^2-m^2} \frac{i}{(q+p_3+k)^2-m^2}$$

The sub-diagram we are interested in corresponds to the integral w.r.t q, namely

$$\begin{aligned}
&\int \frac{d^dq}{(2\pi)^d} \frac{i}{(q+p_3+k)^2-m^2} \frac{i}{q^2-m^2} \\
=&-\int \frac{d^dq}{(2\pi)^d} \frac{1}{(q+p_3+k)^2-m^2} \frac{1}{q^2-m^2}
\end{aligned}$$

Richard Feynman observed

$$\frac{1}{AB} = \int_0^1 dx \frac{1}{(Ax+(1-x)B)^2}$$

For our case

$$\begin{aligned}
A &= (q+p_3+k)^2 - m^2 \\
B &= q^2 - m^2
\end{aligned}$$

Then

$$\begin{aligned}
&\frac{1}{(q+p_3+k)^2-m^2} \frac{1}{q^2-m^2} \\
=&\int_0^1 \frac{1}{\left[x(q+p_3+k)^2 - xm^2 + q^2 - m^2 - xq^2 + xm^2\right]^2} \\
=&\int_0^1 \frac{1}{\left[x(q+p_3+k)^2 - xm^2 + q^2 - m^2 - xq^2 + xm^2\right]^2}
\end{aligned}$$

$$\begin{aligned}
&x(q+p_3+k)^2 - xm^2 + q^2 - m^2 - xq^2 + xm^2 \\
=&xp_3^2 + xk^2 + 2qp_3x + 2qkx + 2p_3kx + q^2 - m^2
\end{aligned}$$

let

$$\begin{aligned}
l &= q + (p_3+k)x \\
l^2 &= (q+(p_3+k)x)^2 \\
&= q^2 + 2(p_3+k)xq + (p_3+k)^2x^2 \\
&= q^2 + \underline{p_3^2x^2} + \underline{k^2x^2} + \underline{2p_3kx^2} + 2p_3xq + 2kxq
\end{aligned}$$

where we need to take away the underlined terms

Then

$$\begin{aligned}
&\int_0^1 \frac{dx}{\left[x(q+p_3+k)^2 - xm^2 + q^2 - m^2 - xq^2 + xm^2\right]^2} \\
=&\int_0^1 \frac{dx}{\left[(q+(p_3+k)x)^2 - x^2(k+p_3)^2 + xp_3^2 + 2p_3kx + xk^2 - m^2\right]^2} \\
=&\int_0^1 \frac{dx}{\left[(q+(p_3+k)x)^2 - x^2(k+p_3)^2 + x(k+p_3)^2 - m^2\right]^2} \\
=&\int_0^1 \frac{dx}{\left[l^2 + x(1-x)(k+p_3)^2 - m^2\right]^2} \\
=&\int_0^1 \frac{dx}{\left[l^2 - \Delta\right]^2}
\end{aligned}$$

where

$$\Delta = m^2 - x(1-x)(k+p_3)^2$$

Thus

$$\begin{aligned} i\mathcal{M} &= (-i\lambda)^3 \int \frac{d^dk}{(2\pi)^d} \int \frac{d^dq}{(2\pi)^d} \frac{i}{k^2-m^2} \frac{i}{(k+p)^2-m^2} \frac{i}{q^2-m^2} \frac{i}{(q+p_3+k)^2-m^2} \\ &= -(-i\lambda)^3 \int \frac{d^dk}{(2\pi)^d} \frac{i}{k^2-m^2} \frac{i}{(k+p)^2-m^2} \int \frac{d^dl}{(2\pi)^d} \int_0^1 dx \frac{1}{[l^2-\Delta]^2} \end{aligned}$$

We can carry out wick rotation $l^0 \to il_E^0$, thus

$$\begin{aligned} &\int_0^1 dx \int \frac{d^dl}{(2\pi)^d} \frac{1}{[l^2-\Delta]^2} \\ =& i \int_0^1 dx \int \frac{d^dl_E}{(2\pi)^d} \frac{1}{[l_E^2+\Delta]^2} \\ =& i \int_0^1 dx \frac{1}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(2-\frac{d}{2})}{\Gamma(2)\Delta^{2-\frac{d}{2}}} = iV_{[\text{1-loop}]} \end{aligned}$$

Thus

$$\begin{aligned} i\mathcal{M} &= (-i\lambda)^3 \int \frac{d^dk}{(2\pi)^d} \frac{i}{k^2-m^2} \frac{i}{(k+p)^2-m^2} iV_{[\text{1-loop}]} \\ &= -i(-i\lambda)^3 \int \frac{d^dk}{(2\pi)^d} \frac{1}{k^2-m^2} \frac{1}{(k+p)^2-m^2} V_{[\text{1-loop}]} \end{aligned}$$

(b) Express $V_{[\text{1-loop}]}$ in terms of the integral over its own loop momentum.

Answer:
We have demonstrated in (a) that

$$\begin{aligned} V_{[\text{1-loop}]} &= \int_0^1 dx \int \frac{d^dl_E}{(2\pi)^d} \frac{1}{[l_E^2+\Delta]^2} \\ &= \int_0^1 dx \frac{1}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(2-\frac{d}{2})}{\Gamma(2)\Delta^{2-\frac{d}{2}}} \end{aligned}$$

(c) Use Feynman parameter representation, separately for $V_{[\text{1-loop}]}$ and $V_{[\text{2-loop}]}$ (which has been already written in (a) in terms of $V_{[\text{1-loop}]}$); do not have to calculate the integrals.

Answer:
We have already carried out Feynman parameter representation for $V_{[\text{1-loop}]}$, not move to the k integral.

$$\int \frac{d^dk}{(2\pi)^d} \frac{i}{k^2-m^2} \frac{i}{(k+p)^2-m^2}$$

Richard Feynman observed

$$\frac{1}{AB} = \int_0^1 dx \frac{1}{(Ax+(1-x)B)^2}$$

For our case

$$\begin{aligned} A &= (k+p)^2 - m^2 \\ B &= k^2 - m^2 \end{aligned}$$

Then

$$\begin{aligned} \frac{1}{k^2-m^2} \frac{1}{(k+p)^2-m^2} &= \int_0^1 dy \frac{1}{(Ay+(1-y)B)^2} \\ &= \int_0^1 dy \frac{1}{(y(k^2+p^2+2kp)-ym^2+(1-y)(k^2-m^2))^2} \\ &= \int_0^1 dy \frac{1}{(yp^2+2kpy+k^2-m^2))^2} \end{aligned}$$

Let

$$
\begin{aligned}
l_y &= k + py \\
l_y^2 &= k^2 + 2pyk + p^2y^2
\end{aligned}
$$

Then

$$
\begin{aligned}
&\frac{1}{(yp^2 + 2kpy + k^2 - m^2))^2} \\
=&\frac{1}{(l_y^2 + yp^2 - p^2y^2 - m^2))^2} \\
=&\frac{1}{(l_y^2 + y(1-y)p^2 - m^2)^2} \\
=&\frac{1}{(l_y^2 - \Delta_y)^2}
\end{aligned}
$$

where

$$
\Delta_y = m^2 - y(1-y)p^2
$$

Thus

$$
\begin{aligned}
\frac{1}{k^2 - m^2}\frac{1}{(k+p)^2 - m^2} &= \int_0^1 dy \frac{1}{(yp^2 + 2kpy + k^2 - m^2))^2} \\
&= \int_0^1 dy \frac{1}{(l^2 - \Delta_y)^2}
\end{aligned}
$$

Then

$$
\begin{aligned}
i\mathcal{M} &= (-i\lambda)^3 \int \frac{d^d l_y}{(2\pi)^d} \frac{i}{k^2 - m^2} \frac{i}{(k+p)^2 - m^2} iV_{[\text{1-loop}]} \\
&= i(-i\lambda)^3 \int \frac{d^d l_y}{(2\pi)^d} \int_0^1 dy \frac{1}{(l_y^2 - \Delta_y)^2} V_{[\text{1-loop}]}
\end{aligned}
$$

(d) Is this 2-loop diagram divergent or convergent? Explain the answer.

Answer:
We end up with

$$
i\mathcal{M} = i(-i\lambda)^3 \int \frac{d^d l}{(2\pi)^d} \int_0^1 dy \frac{1}{(l^2 - \Delta_y)^2} V_{[\text{1-loop}]}
$$

Via dimensional regularization we can demonstrate when $d = 4$, the above integral diverges logarithmically w.r.t Δ_y.